FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – IV

JEE (Main)-2022

<u>TEST DATE: 04-07-2022</u>

Time Allotted: 3 Hours **Maximum Marks: 300**

<u>General Instructions:</u>

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 Questions.

Section-A (01 – 20, 31 – 50, 61 – 80) *contains 60 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.*

***Section-B (21 – 30, 51 – 60, 81 – 90)** contains 30 Numerical answer type questions with answer XXXXX.XX and each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.*

Physics

PART – A

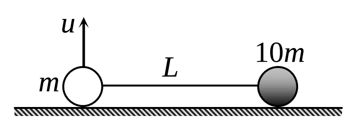
SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A stone is dropped from a height *h*. Simultaneously, another stone is thrown up from the ground which reaches a height 4 *h*. The two stones cross each other after time

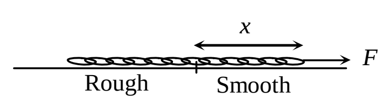
(A) $\sqrt{\frac{h}{2g}}$
(B) $\sqrt{\frac{h}{8g}}$
(C) $\sqrt{2hg}$
(D) $\sqrt{8hg}$

2. As shown in the figure, a mass *m* and another mass 10*m* are connected with a string. Friction is sufficient to prevent the slipping of 10*m*. Mass *m* is given a velocity *u* in vertical direction. For complete circular motion of mass *m* keeping heavier particle stationary, the value of *u* is



(A) $u > \sqrt{3gL}$
(B) $\sqrt{3gL} < u < \sqrt{5gL}$
(C) $\sqrt{3gL} < u < \sqrt{13gL}$
(D) $\sqrt{11gL} < u < \sqrt{13gL}$

3. A chain of mass per unit length $\lambda = 2$ kg/m is pulled up by a constant force *F*. Initially the chain is lying on rough surface and passes onto the smooth surface. The co-efficient of friction between chain and rough surface is $\mu = 0.1$. The length of the chain is *L*. The velocity of the chain when $x = L$ is

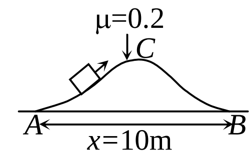


(A) $\sqrt{F-L}$
(B) $\sqrt{F-2L}$
(C) $\sqrt{F-4L}$
(D) $\sqrt{F-\frac{L}{2}}$

4. A body of mass m starting from rest moves with acceleration proportional to cube root of its velocity. The instantaneous power delivered to the body is varying with time as

(A) t^2
(B) $t^{2/3}$
(C) $t^{4/3}$
(D) $t^{9/8}$

5. A block of mass 1 kg is pulled along the curve path *ACB* by a tangential force as shown in figure. The work done by the frictional force when the block moves from *A* to *B* is

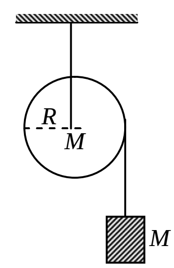


(A) 5 J
(B) 10 J
(C) 20 J
(D) none of these

6. A small ball thrown at an initial velocity *u* directed at an angle θ = 37° above the horizontal collides inelastically (*e* = 1/4) with a vertical massive wall moving with a uniform horizontal velocity *u*/5 towards ball. After collision with the wall, the ball returns to the point from where it was thrown. Neglect friction between ball and wall. The time *t* from beginning of motion of the ball till the moment of its impact with the wall is (tan37° = 3/4)

(A) $\frac{3u}{5g}$
(B) $\frac{18u}{25g}$
(C) $\frac{54u}{125g}$
(D) $\frac{54u}{25g}$

7. A mass *M* is supported by a massless string wound round a uniform cylinder of mass *M* and radius *R*. On releasing the mass from rest, it will fall with acceleration :



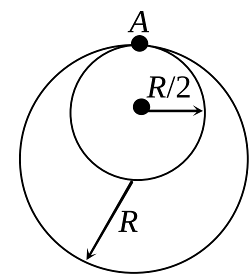
(A) g
(B) $\frac{1}{2}g$
(C) $\frac{1}{3}g$
(D) $\frac{2}{3}g$

8. A body is projected up with a velocity equal to $\frac{3}{4}$ of the escape velocity from the surface of the earth. The height it reaches is: (Radius of the earth = *R*)

(A) $\frac{10R}{9}$
(B) $\frac{9R}{7}$
(C) $\frac{9R}{8}$
(D) $\frac{10R}{3}$

9. A cone made of a material of relative density $\left(s=\dfrac{27}{64}\right)$ and height 4 m floats with its apex downward in a big vessel containing water. The submerged height of cone in water is
(A) 3 m
(B) 2 m
(C) 3.5 m
(D) 2.5 m

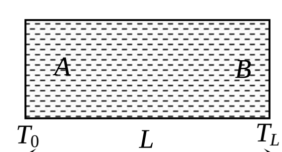
10. Two rings made of same material and thickness, one of radius R and other of radius $R/2$ are welded together at point A. Now, it is hanged on a nail at wall, the nail touching both the rings at A. Now it is slightly displaced in the plane of rings and released. The period of small oscillations is



(A) $2\pi\sqrt{\dfrac{2R}{5g}}$
(B) $2\pi\sqrt{\dfrac{5R}{6g}}$
(C) $2\pi\sqrt{\dfrac{9R}{5g}}$
(D) $2\pi\sqrt{\dfrac{5R}{2g}}$

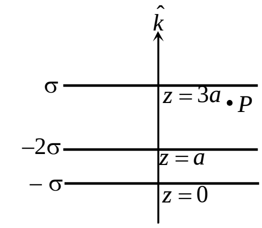
11. A 3.6 m long vertical pipe is filled completely with a liquid. A small hole is drilled at the base of the pipe due to which liquids starts leaking out. This pipe resonates with a tuning fork. The first two resonances occur when height of water column is 3.22 m and 2.34 m respectively. The area of cross-section of pipe is
(A) $25\,\pi$ cm^2
(B) $100\,\pi$ cm^2
(C) $200\,\pi$ cm^2
(D) $400\,\pi$ cm^2

12. The temperature of a mono-atomic gas in an uniform container of length 'L' varies linearly from T_0 to T_L as shown in the figure. If the molecular weight of the gas is M, then the time taken by a wave pulse in traveling from end A to end B is



(A) $\dfrac{2L}{\sqrt{T_L}+\sqrt{T_0}}\sqrt{\dfrac{3M}{5R}}$
(B) $\sqrt{\dfrac{3(T_L-T_0)}{5RML}}$
(C) $\dfrac{2L}{\sqrt{T_L}-\sqrt{T_0}}\sqrt{\dfrac{3M}{5R}}$
(D) $L\sqrt{\dfrac{M}{2R(T_L-T_0)}}$

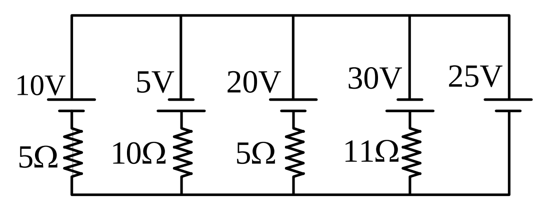
13. Three large thin sheets with surface charge density σ, –2σ and –σ are placed as shown in figure. Electric field intensity at the point *P* is



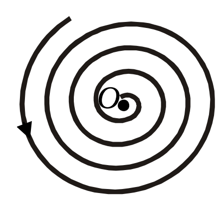
(A) $-\frac{4\sigma}{\varepsilon_0}\hat{k}$
(B) $\frac{4\sigma}{\varepsilon_0}\hat{k}$
(C) $-\frac{2\sigma}{\varepsilon_0}\hat{k}$
(D) $\frac{2\sigma}{\varepsilon_0}\hat{k}$

14. A ball of mass 2 kg having charge 1 μC is dropped from the top of a high tower. In space electric field exist in horizontal direction away from tower which varies as $E = (5 - 2x) \times 10^6$ V/m (where *x*-is horizontal distance from tower), the maximum horizontal distance ball can go from the tower is
(A) 5m
(B) 2.5m
(C) 10m
(D) 15m

15. In the circuit shown, current through 25V cell is
(A) 7.2 A
(B) 10 A
(C) 12 A
(D) 14.2 A

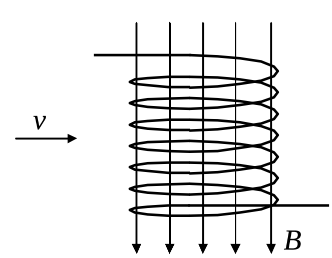


16. A positive charged particle is moving in a plane spiral path with constant angular velocity and constant centripetal velocity $\left(-\frac{dr}{dt}\right)$. If a current carrying wire is placed at *O* perpendicular to the plane of spiral path, then path followed by the charged particle is



(A) helical path with constant radius and constant pitch
(B) helical path with decreasing radius and increasing pitch
(C) helical path with decreasing radius and decreasing pitch
(D) path will remain same

17. A direct current flowing through the winding of a long cylindrical solenoid of radius *R* produces in it a uniform magnetic field of induction *B.* An electron flies into the solenoid along the radius between its turns (at right angles to the solenoid axis) at a velocity *v* as shown in figure. After a certain time, the electron deflected by the magnetic field leaves the solenoid. Then the time *t* during which the electron moves in the solenoid is



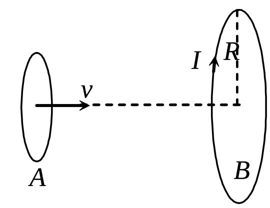
(A) $\frac{m}{eB}\tan^{-1}\left(\frac{eBR}{mv}\right)$

(B) $\frac{2m}{eB}\tan^{-1}\left(\frac{eBR}{mv}\right)$

(C) $\frac{m}{eB}\tan^{-1}\left(\frac{mv}{eBR}\right)$

(D) $\frac{2m}{eB}\tan^{-1}\left(\frac{mv}{eBR}\right)$

18. Loop *A* of radius $r(r << R)$ moves towards a constant current carrying loop *B* with a constant velocity *v* in such a way that their planes are parallel and coaxial. The distance between the loops when the induced emf in loop *A* is maximum is



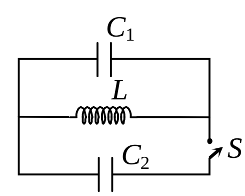
(A) *R*

(B) $\frac{R}{\sqrt{2}}$

(C) $\frac{R}{2}$

(D) $R\left(1-\frac{1}{\sqrt{2}}\right)$

19. At a moment (t = 0), when the charge on capacitor C_1 is zero, the switch is closed. If I_0 be the current through inductor at t = 0 , for t > 0



(A) maximum current through inductor equals $I_0/2$.

(B) maximum current through inductor equals $\frac{C_1 I_0}{C_1+C_2}$.

(C) maximum charge on $C_1 = \frac{C_1 I_0\sqrt{LC_2}}{C_1+C_2}$.

(D) maximum charge on $C_1 = C_1 I_0\sqrt{\frac{L}{C_1+C_2}}$.

20. At a point on the screen in YDSE, 3rd maxima is observed at t = 0. Now screen is slowly moved with constant speed away from the slits in such a way that the centre of slits and centre of screen lie on same line always and at t = 1s, the intensity at that point is observed to be $\left(\frac{3}{4}\right)^{\text{th}}$ of maximum intensity in between 2nd and 3rd maxima. The speed of screen will be (D = 1m)

(A) $\frac{5}{13}$ m/s

(B) $\frac{5}{12}$ m/s

(C) $\frac{12}{5}$ m/s

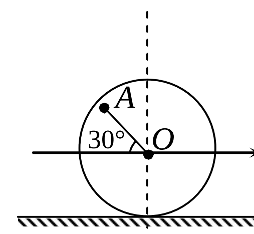
(D) $\frac{13}{5}$ m/s

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

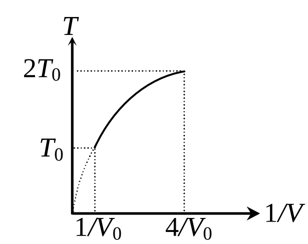
21. Two bodies of mass 1 kg and 2 kg move towards each other in mutually perpendicular direction with the velocities 3 m/s and 2 m/s respectively. If the bodies stick together after collision the heat liberated will be

22. The wheel of radius $r = 300$ mm rolls to the right without slipping and has a velocity $v_0 = 3$ m/s of its center *O*. The speed of the point *A* on the wheel for the instant represented in the figure is (*OA* = 20 mm, $\sqrt{17} = 4.36$)

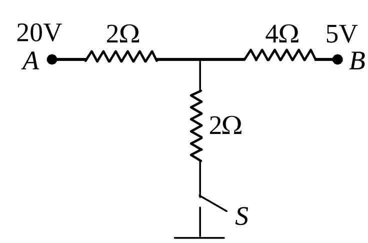


23. A cylindrical tank has a hole of 1 cm^2 in its bottom. If the water is allowed to flow into the tank from a tube above it at the rate of 70 cm^3/sec then the maximum height (in cm) up to which water can rise in the tank is

24. Figure shows a parabolic graph between *T* and $\frac{1}{V}$ (*T* = temperature, *V* = volume) for a mixture of a gases undergoing an adiabatic process. The ratio of rms velocity of molecules and speed of sound in the mixture of gases at the same temperature is



25. As the switch *S* is closed in the circuit shown in figure current passed through it is



26. The *x*-*z* plane separates two media *A* and *B* of refractive indices μ_1 = 1.5 and μ_2 =2. A ray of light travels from *A* to *B*. Its directions in the two media are given by unit vectors $\vec{u}_1 = a\hat{i} + b\hat{j}$ and $\vec{u}_2 = c\hat{i} + d\hat{j}$. Then

27. Electromagnetic radiation whose electric component varies with time as $E = C_1(C_2 + C_3 \cos \omega t)\cos \omega_0 t$, here C_1, C_2 and C_3 are constants, is incident on lithium and librates photoelectrons. If the kinetic energy of most energetic electrons be 0.592 × 10^{-19} J. Given that ω_0 = 3.6 × 10^{15} rad/sec and ω = 6 × 10^{14} rad/sec. The work function (in ev) of lithium is (take planks constant h = 6.6 × 10^{-34} MKS).

28. The wavelength of the characteristic X-ray K_α line emitted by a hydrogen-like element is 0.32 Å. The wavelength of K_β line (in Å) emitted by the same element will be

29. What is the conductivity of a semiconductor if electron density is 5 × 10^{12} per cm^3 and hole density is 8 × 10^{13} per cm^3. (μ_e = 2.3 V^{-1} s^{-1} m^2 and μ_h = 0.01 $V^{-1}s^{-1}m^2$)

30. The pitch of a screw gauge is 1 mm and there are 100 division on its circular scale. When nothing is put in between its jaws, the zero of the circular scale lies 4 divisions below the reference line. When a steel wire is placed between the jaws, two main scale divisions are clearly visible and 67 divisions on the circular scale are observed. The diameter of the wire (in mm) is

Chemistry

PART – B

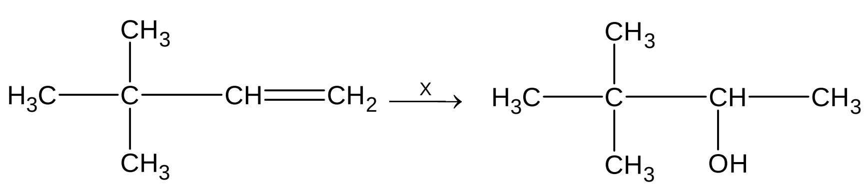
SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. DNA molecule is formed of:
(A) pentose sugar, pyrimidines and purines
(B) pentose sugar, phosphoric acid, pyrimidines and purines
(C) pentose sugar, phosphoric acid and purines
(D) chloridepentose sugar, phosphoric acid and pyrimidines

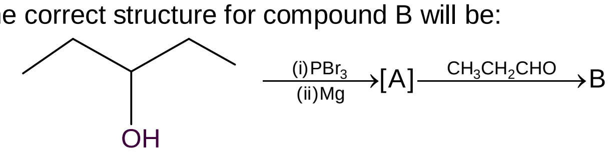
32. Which of the following esters cannot undergo Claisen self condensation?
(A) $CH_3CH_2CH_2CH_2COOC_2H_5$
(B) $C_6H_5COOC_2H_5$
(C) $C_6H_{11}CH_2COOC_2H_5$
(D) $C_6H_5CH_2COOC_2H_5$

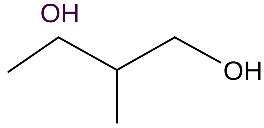
33. Two isomeric ketones, 3-pentanone and 2-pentanone can be distinguished by:
(A) I_2/NaOH only
(B) $NaSO_3H$ only
(C) NaCN/HCl
(D) Both (A) and (B)

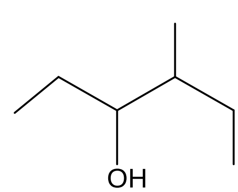
34. 

X can be:
(A) BH_3 / THF,H_2O_2 / OH^-
(B) H_3O^+
(C) $Hg(OAc)_2$ / $NaBH_4$,NaOH
(D) All of these

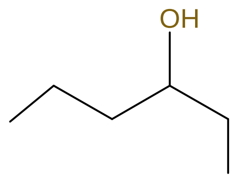
35. The correct structure for compound B will be:



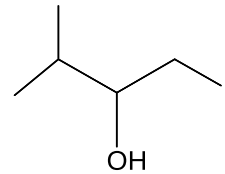
(A) 

(B) 

(C)



(D)

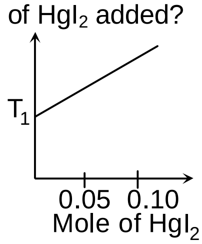


36. Hydroxylamine reduces iron (III) according to following equation
$NH_2OH + Fe_2(SO_4)_3 \rightarrow N_2(g) + H_2O + FeSO_4 + H_2SO_4$
Which statement is correct?
(A) n-factor for hydroxylamine is 1.
(B) equivalent weight of $Fe_2(SO_4)_3$ is M/2
(C) 6 milliequiv. of $Fe_2(SO_4)_3$ is contained in 3 millimoles of ferric sulphate.
(D) All of these

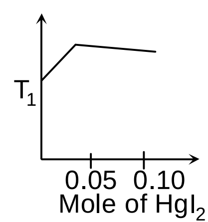
37. For a saturated solution of AgCl at 25^{o}C, specific conductance is 3.41×10^{-6} ohm^{-1}cm^{-1} and that of water used for preparing the solution was $1.60 \times 10^{-6}\ ohm^{-1}\,cm^{-1}$. What is the solubility product of AgCl?
$\wedge_{eq}^{\infty}(AgCl) = 138.3\,ohm^{-1}\,cm^{-1}\,equiv^{-1}$.
(A) $1.72 \times 10^{-10}\ mol^2L^{-2}$
(B) $1.3 \times 10^{-8}\ mol^2L^{-2}$
(C) $2.5 \times 10^{-8}\ mol^2L^{-2}$
(D) $1.8 \times 10^{-9}\ mol^2L^{-2}$

38. Increasing amount of solid HgI_2 is added to 1L of an aqueous solution containing 0.1 mol KI. Which of the following graphs represents the variation of freezing point of the resulting with the amount of HgI_2 added?

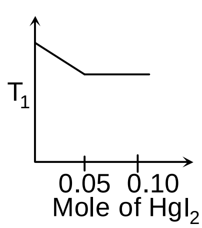
(A)



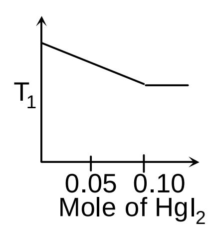
(B)



(C)

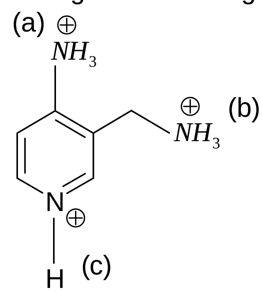


(D)



39. The rate constant of a certain first order reaction increases by 11.11% per degree rise of temperature at 27^o. By what % will it increase at 127^{o}C, assuming constancy of activation energy over the given temperature range?
(A) 5.26 %
(B) 5.62 %
(C) 6.25 %
(D) 7.33 %

40. Arrange the following marked species in decreasing order of Acidic strength



(A) a>c>b
(B) a>b>c
(C) c>a>b
(D) c>b>a

41. The maximum orbital angular momentum of an electron with n = 5 is

(A) $\sqrt{6}\frac{h}{2\pi}$

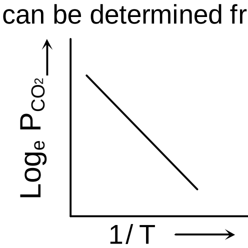
(B) $\sqrt{12}\frac{h}{2\pi}$

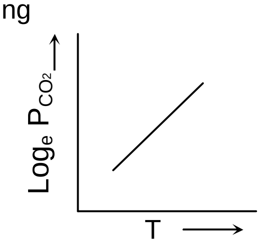
(C) $\sqrt{42}\frac{h}{2\pi}$

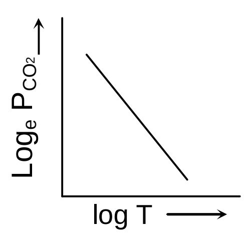
(D) $\sqrt{20}\frac{h}{2\pi}$

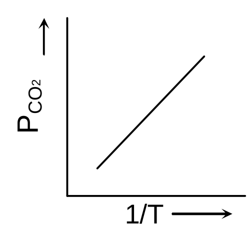
42. For the chemical equilibrium,
$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

$\Delta_r H^o$ can be determined from which one of the following

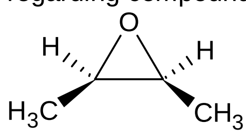
(A)


(B)


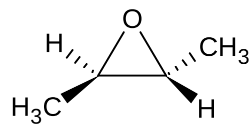
(C)


(D)


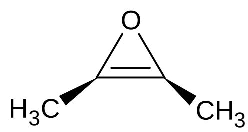
43. Epoxides are three membered rings containing one oxygen atom with torisonal strain energies and angle strain energies similar to cyclopropanes. Compound I is less stable than compound II but compound II is more stable than compound III. Which of the following statements is true regarding compounds I, II and III.



(I)



(II)



(III)

(A) I has more torisonal strain than II
(B) I and II have equal angle strain
(C) The angle strain energy of III is less than the total torisonal strain energy and angle strain energy of II.
(D) Both (A) and (B) are true.

44. $Ag_2S + NaCN \rightarrow [A]$

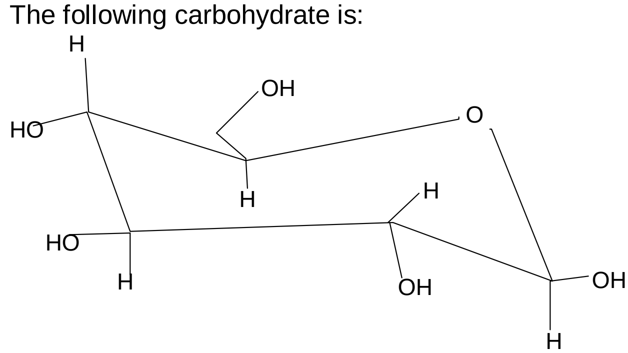
$$[A] + Zn \rightarrow [B]$$

[B] is a metal. Hence, [A] and [B] are:
(A) $Na_2[Zn(CN)_4], Zn$
(B) $Na[Ag(CN)_2], Ag$
(C) $Na_2[Ag(CN)_4], Ag$
(D) $Na_3[Ag(CN)_4], Ag$

45. The rate constant, the activation energy and the arrhenius parameters of a chemical reaction at 25^oC are $2.0\times10^{-5}\,s^{-1}, 100\,kJ\,mol^{-1}$ and $6.0\times10^{14}\,s^{-1}$, respectively. The value of rate constant at very high temperature approaches.
(A) $2.0\times10^{-5}\,s^{-1}$
(B) infinity
(C) $6.0\times10^{14}\,s^{-1}$
(D) $12\times10^{-9}\,s^{-1}$

46. If the pressure in a reaction vessel for the following reaction is increased by decreasing the volume, what will happen to the concentrations of CO and CO_2?
$H_2O(g) + CO(g) \rightleftharpoons H_2(g) + CO_2(g) + \text{Heat}$
(A) both the [CO] and [CO_2] will decrease
(B) neither the [CO] nor the [CO_2] will change
(C) the [CO] will decrease and the [CO_2] will increase
(D) both the [CO] and [CO_2] will increase

47. The following carbohydrate is:



(A) a ketohexose
(B) an aldohexose
(C) an α-furanose
(D) an α-pyranose

48. Which of the following has two nodal planes?
(A) σ^*_{ns}
(B) $\sigma^*_{np_z}$
(C) $\pi^*_{2p_x}$
(D) π_{2p_x}

49. When N_2O_5 is heated at certain temperature, it dissociates as
$N_2O_5(g) \rightleftharpoons N_2O_3(g) + O_2(g); K_c = 2.5$. At the same time N_2O_3 also decomposes as:
$N_2O_3(g) \rightleftharpoons N_2O(g) + O_2(g)$. If initially 4.0 moles of N_2O_5 are taken in 1.0 litre flask and allowed to dissociation, concentration of O_2 at equilibrium is 2.5M. Equilibrium concentration of N_2O_5 is:
(A) 1.0 M
(B) 1.5 M
(C) 2.166 M
(D) 1.846 M

50. The basic component of smog is:
(A) PAN
(B) PBN
(C) NO_2
(D) All of these

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

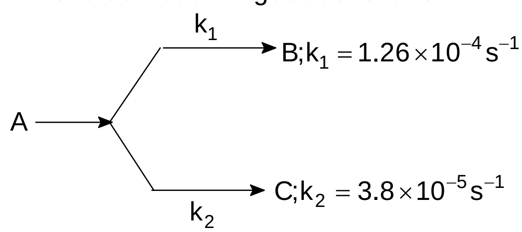
51. At 25°C, when 1 mole of $MgSO_4$ was dissolved in water, the heat evolved was found to be 91.2 kJ. One mole of $MgSO_4.7H_2O$ on dissolution gives a solution of the same composition accompanied by an absorption of 13.8 kJ. The enthalpy of hydration, i.e., $\frac{\Delta H}{10}$ for the reaction (in kJ/mol)
$MgSO_4(s) + 7H_2O(l) \rightarrow MgSO_4.7H_2O(s)$ is :

52. An aqueous solution contains 3% and 1.8% by mass, Urea and glucose respectively. What is the freezing point of solution? ($K_f = 1.86^oC/m$)

53. First and second ionisation energies of Mg(g) are 740 and 1450 kJ mol^{-1}. Calculate percentage of $Mg^+(g)$ if 1 g of Mg(g) absorbs 50 kJ of energy.

54. $Cu^+(aq)$ is unstable in solution and undergoes simultaneous oxidation and reduction according to the reaction:
$2Cu^+ \rightleftharpoons Cu^{2+}(aq) + Cu(s)$
Choose the correct E^o for the reaction. If $E^o_{Cu^{2+}/Cu} = 0.34$ and $E^o_{Cu^{2+}/Cu^+} = 0.15V$.

55. A first order reaction goes as follows:



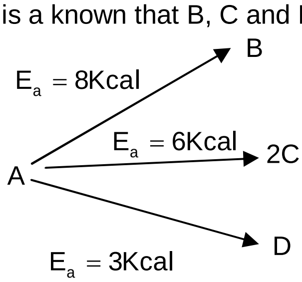
The percentage of B in the mixture of B and C is likely to be:

56. The coagulation of 200 mL of standard red gold sol is completely prevented by the addition of 0.44g of potato starch to it when every 10 mL of this gold sol was treated with 1 mL of 10% NaCl. The coagulation number of potato starch is:

57. What is the concentration of glucose solution in gL^{-1} that is isotonic with human blood having osmotic pressure 8.2 atm at 37^oC?
(mol. wt. of glucose = 180g, R=0.082L atm K^{-1} mol^{-1})

58. EMF of the following cell is 0.67 V at 298 K.
$Pt|H_2(1atm)|H^+(pH = X)||KCl(1N)|Hg_2Cl_2(s)|Hg$
Calculate pH of the anode compartment. Given: $E_{Cl^-|Hg_2Cl_2|Hg} = 0.28V$.

59. A substance undergoes first order parallel reaction as shown. Calculate net activation energy if it is a known that B, C and D are formed in a molar ratio of 2: 3: 4 respectively



60. In a sample of three H-atom, all in the 5^{th} excited state, electrons make a transition to 1^{st} excited state. The maximum number of different spectral lines observed will be:

Mathematics

PART – C

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. A straight line passes through the point (2, –1, –1). It is parallel to the plane $4x + y + z + 2 = 0$ and is perpendicular to the line $\frac{x}{1} = \frac{y}{-2} = \frac{z-5}{1}$. The equations of the straight line are

(A) $\frac{x-2}{4} = \frac{y+1}{1} = \frac{z+1}{1}$

(B) $\frac{x+2}{4} = \frac{y-1}{1} = \frac{z-1}{1}$

(C) $\frac{x-2}{-1} = \frac{y+1}{1} = \frac{z+1}{3}$

(D) $\frac{x+2}{-1} = \frac{y-1}{1} = \frac{z-1}{3}$

62. If a, b, c, d, e and f are non negative real numbers such that a + b + c +d + e + f = 1, then the maximum value of ab + bc + cd + de + ef is

(A) $\frac{1}{6}$

(B) 1

(C) 6

(D) $\frac{1}{4}$

63. If the area bounded by the curve y = f(x), (f(x) > 0, $\forall$ x $\in R^{+}$) the x axis, the y axis and the line $x = a$ is $\frac{6a - \sin 2a}{4}$ $\forall$a>0, then f(x) is

(A) $1 + \sin^2 x, x > 0$

(B) $1-\cos^2 x$, x > 0

(C) $\frac{1+\cos^2 x}{4}$, x > 0

(D) $\frac{1-\cos^2 x}{4}, x > 0$

64. Let n be an odd natural number and A = $\sum_{r=1}^{\frac{n-1}{2}} \frac{1}{{}^nC_r}$. Then value of $\sum_{r=1}^{n} \frac{r}{{}^nC_r}$ is equal to

(A) n(A-1)

(B) n(A+1)

(C) $\frac{nA}{2}$

(D) nA

65. The chords of contact of the pair of tangents drawn from each point on the line $2x + y = 4$ to the circle $x^2 + y^2 = 1$ pass through fixed point
(A) (2, 4)
(B) $\left(-\frac{1}{2}, -\frac{1}{4}\right)$
(C) $\left(\frac{1}{2}, \frac{1}{4}\right)$
(D) (-2, -4)

66. The value of $i \log(x - i) + i^2\pi + i^3 \log(x + i) + i^4(2 \tan^{-1}x)$, $x > 0$ (where $i = \sqrt{-1}$) is
(A) 0
(B) 1
(C) 2
(D) 3

67. The equation of the curve, passing through (2, 5) and having the area of triangle formed by the x-axis, the ordinate of a point on the curve and the tangent at the point 5 sq units, is
(A) $xy = 10$
(B) $x^2 = 10y$
(C) $y^2 = 10x$
(D) $xy^{1/2} = 10$

68. The determinant $\Delta(x) = \begin{vmatrix} a^2(1+x) & ab & ac \\ ab & b^2(1+x) & bc \\ ac & bc & c^2(1+x) \end{vmatrix}$ ($abc \neq 0$) is divisible by
(A) $1 + x$
(B) $(1 + x)^2$
(C) x^2
(D) none of these

69. If $I(n) = \int_0^{\pi/2} \theta . \sin^n \theta d\theta$, $n \in N$, $n > 3$, then the value of $I(n) - \frac{n-1}{n} I(n-2)$ is
(A) $\frac{1}{n^2}$
(B) $\frac{1}{n-1}$
(C) $\frac{1}{n^2-1}$
(D) $\frac{1}{n+1}$

70. The area of the triangle formed by the points on the ellipse $25x^2 + 16y^2 = 400$ whose eccentric angles are $\pi/2$, π and $3\pi/2$ is
(A) 10 sq. units
(B) 20 sq. units
(C) 30 sq. units
(D) 40 sq. units

71. If the derivative of f(x) w.r.t. x is $\dfrac{\frac{1}{2}-\sin^2 x}{f(x)}$, then f(x) is a periodic function with period
(A) π
(B) 2π
(C) $\pi/2$
(D) none of these

72. A falg staff on the top of a house subtends the same angle β at two points with distances a and b from the house and on the same side of it. The length of the flag staff is
(A) $(a + b)\sin\beta$
(B) $(a + b)\cos\beta$
(C) $(a + b)\tan\beta$
(D) $(a + b)\cot\beta$

73. If tangent at a point on the hyperbola $\dfrac{x^2}{a^2}-\dfrac{y^2}{b^2}$ = 1 cuts the axis at A and B and rectangle OAPB is completed, then locus of point P is given by (when O is the centre of the hyperbola)
(A) $\dfrac{a^2}{x^2}-\dfrac{b^2}{y^2}=1$
(B) $\dfrac{a^2}{y^2}+\dfrac{b^2}{x^2}=1$
(C) $\dfrac{a^2}{y^2}-\dfrac{b^2}{x^2}=1$
(D) $\dfrac{x^2}{a\sqrt{1-b^2}}+\dfrac{y^2}{b\sqrt{1-a^2}}=1$

74. If $\int \dfrac{x+\left(\cos^{-1}3x\right)^2}{\sqrt{1-9x^2}}dx = A\sqrt{1-9x^2}+B\left(\cos^{-1}3x\right)^3+c$, where c is integration constant, then the value of A and B are,
(A) $A=-\dfrac{1}{9}$; $B=-\dfrac{1}{9}$
(B) $A=-\dfrac{1}{9}$; $B=\dfrac{1}{9}$
(C) $A=\dfrac{1}{9}$; $B=\dfrac{1}{9}$
(D) none of these

75. If f(x) is a continuous function $\forall$ x $\in$ R and the range of f(x) = (2, $\sqrt{26}$) and g(x) = $\left[\dfrac{f(x)}{a}\right]$ is continuous $\forall$ x $\in$ R ([.] denotes the greatest integer function), then the least positive integral value of a is
(A) 2
(B) 3
(C) 6
(D) 5

76. The number of permutations of the letters of the word HINDUSTAN such that neither the pattern 'HIN' nor 'DUS' nor 'TAN' appears, are
(A) 166674
(B) 169194
(C) 166680
(D) 181434

77. If one end of the diameter of a circle is (3, 4) which touches the x-axis then the locus of other end of the diameter of the circle is
(A) parabola
(B) hyperbola
(C) ellipse
(D) none of thee

78. A die is thrown a fixed number of times. If probability of getting even number 3 times is same as the probability of getting even number 4 times, then probability of getting even number exactly once is
(A) $\frac{1}{4}$
(B) $\frac{3}{128}$
(C) $\frac{5}{64}$
(D) $\frac{7}{128}$

79. If $\frac{1}{\sqrt{b}+\sqrt{c}}, \frac{1}{\sqrt{c}+\sqrt{a}}, \frac{1}{\sqrt{a}+\sqrt{b}}$ are in A.P. then 9^{ax+1}, 9^{bx+1}, 9^{cx+1}, $x \neq 0$ are in
(A) G.P.
(B) G.P. only if $x < 0$
(C) G.P. only if $x > 0$
(D) none of these

80. Consider the equation $x^2 + x - n = 0$, where n is an integer lying between 1 to 100. Total number of different values of 'n' so that the equation has integral roots is
(A) 6
(B) 4
(C) 9
(D) None of these

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

81. Value of $\cos(2\cos^{-1}(4/5))$ equals to

82. The sum of the factors of 7!, which are odd and are of the form 3t + 1 where t is a whole number, is

83. Number of real normals drawn from the point (4, 2) to the parabola $y^2 = 8x$ is

84. The probability of having at least one tail in 4 throws with a coin is

85. If $x_i > 0$, i = 1, 2, …., 50 and $x_1 + x_2 + \ldots + x_{50} = 50$, then the minimum value of $\frac{1}{x_1} + \frac{1}{x_2} + \ldots + \frac{1}{x_{50}}$ equals to

86. If a, b, c be the sides of ΔABC and equations $ax^2 + bx + c = 0$ and $5x^2 + 12x + 13 = 0$ have a common root, then $\angle C$ is

87. Area of the quadrilateral formed by lines $|x| + |y| = 1$ is

88. The value of $\cot^2 36^\circ \cot^2 72^\circ$ is

89. Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = x_1\hat{i} + x_2\hat{j} + x_3\hat{k}$, where $x_1, x_2, x_3 \in \{-3, -2, -1, 0, 1, 2\}$.
Number of possible vectors $\vec{b}$ such that $\vec{a}$ and $\vec{b}$ are mutually perpendicular, is

90. Let $\vec{a}$ and $\vec{b}$ be two unit vectors such that $\vec{a} + \vec{b}$ is also a unit vector. The angle between $\vec{a}$ and $\vec{b}$ is